

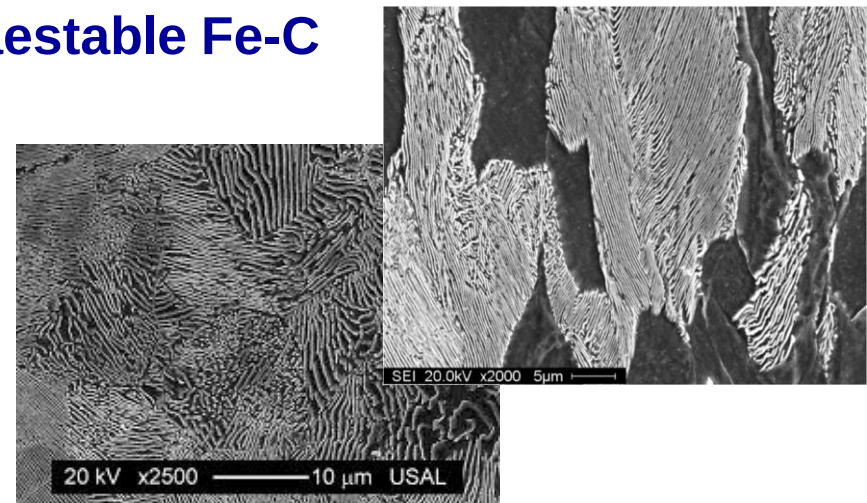
TOPIC 4. Phase Diagrams (III)

1. Fe-C System

2. Equilibrium transformations in the metastable Fe-C system

- Eutectoid steels
- Pearlitic transformations
- Hypoeutectoid steels
- Hypereutectoid steels

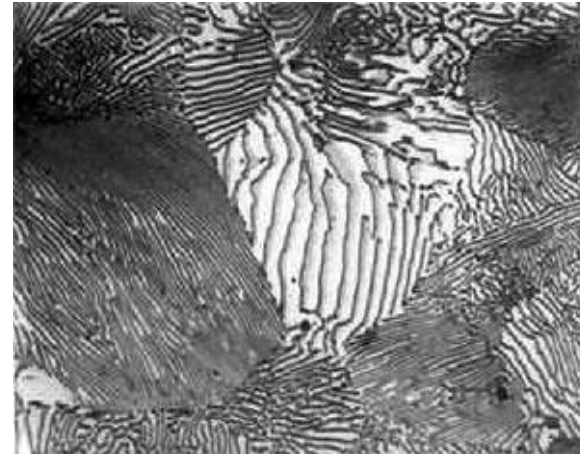
3. Influence of alloy elements in metaestable Fe-C



Fe-C SYSTEM

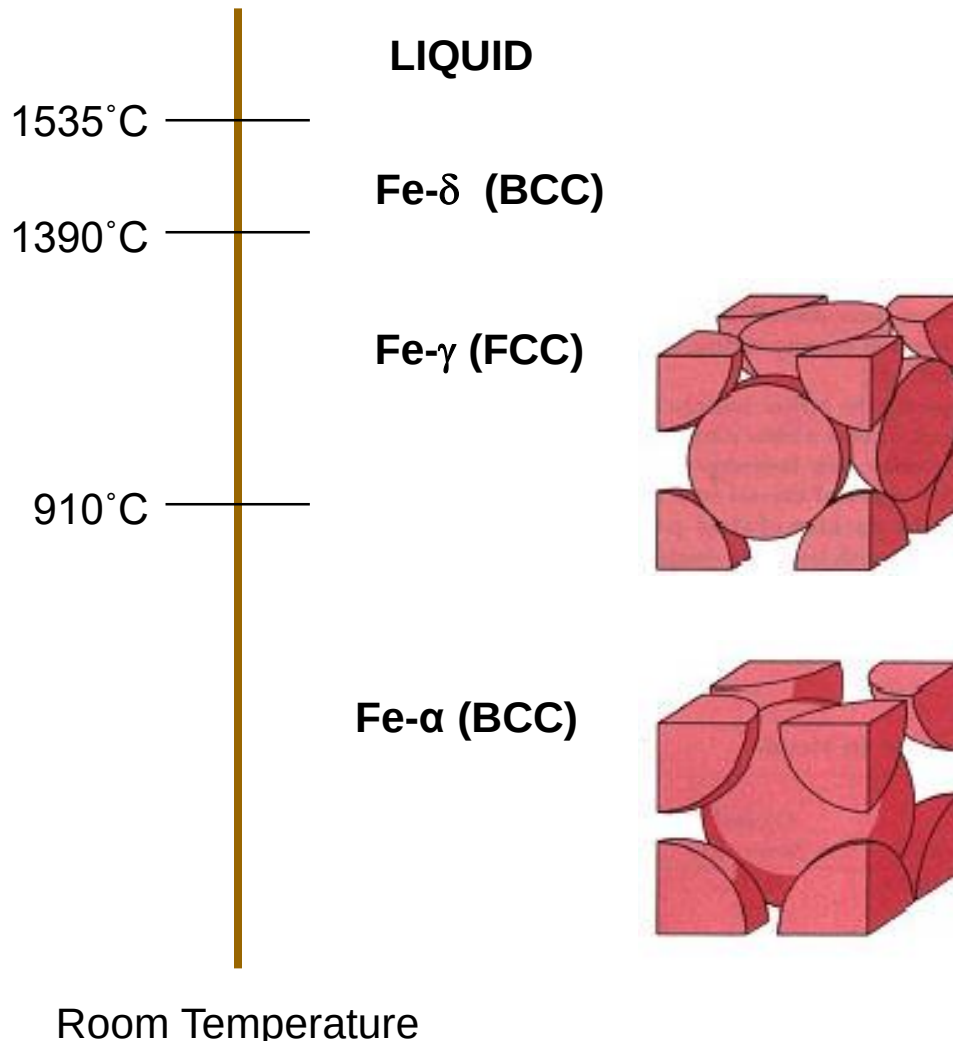
The Fe-C system

- **Fe-C system** is a binary alloy system composed of iron & carbon
- **The most important** alloy system for this civilization: steel & cast iron
- Fe-C system is a **binary eutectic system**
- Depending on composition of carbon, temperature, cooling rate, pressure etc., **different microstructures** can be developed with different properties and applications

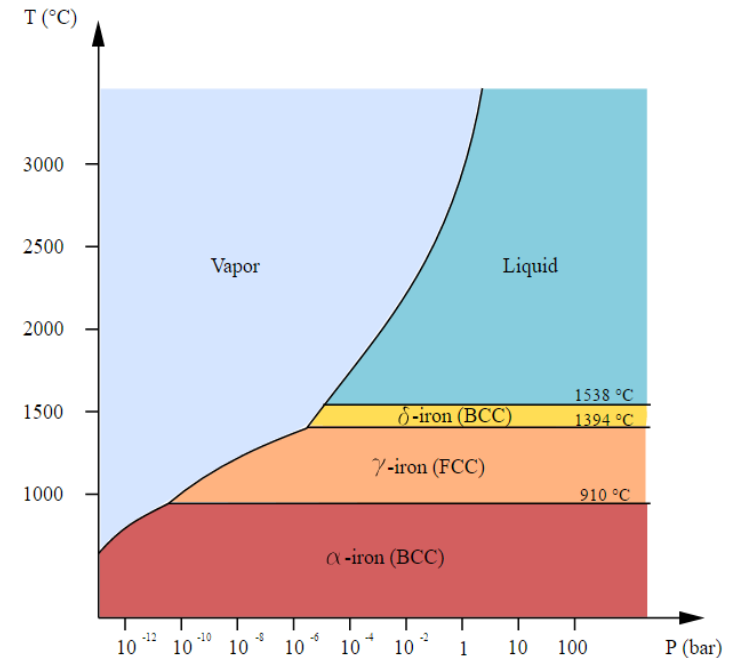


Fe-C SYSTEM

Allotropic phases of pure iron



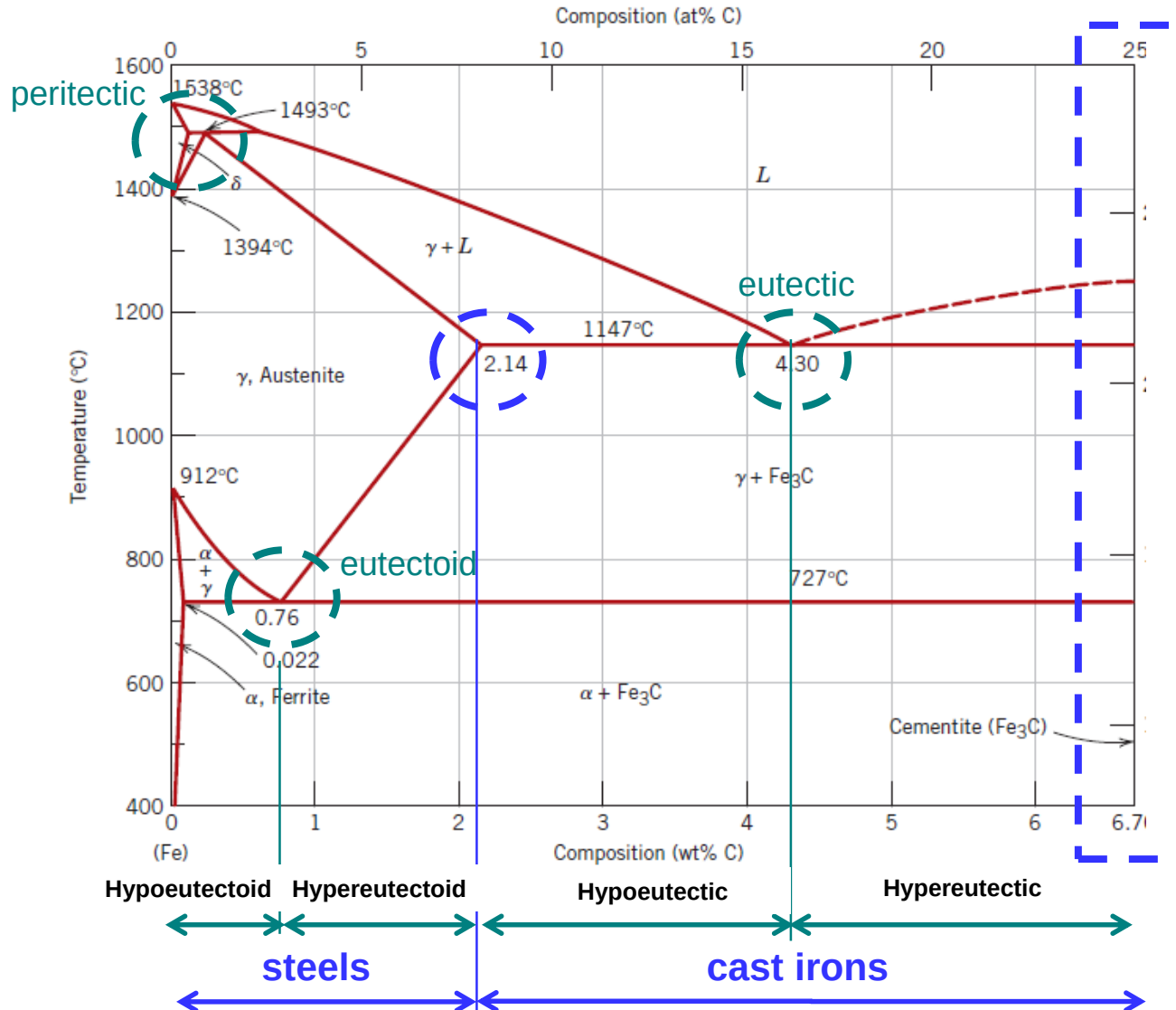
Allotropy: existence of a chemical element in two or more forms, which may differ in the arrangement of atoms in crystalline solids.



Fe experiences two changes in crystal structure before it melts.

Fe-C SYSTEM

Phase diagram of the Fe-C system



Metastable diagram

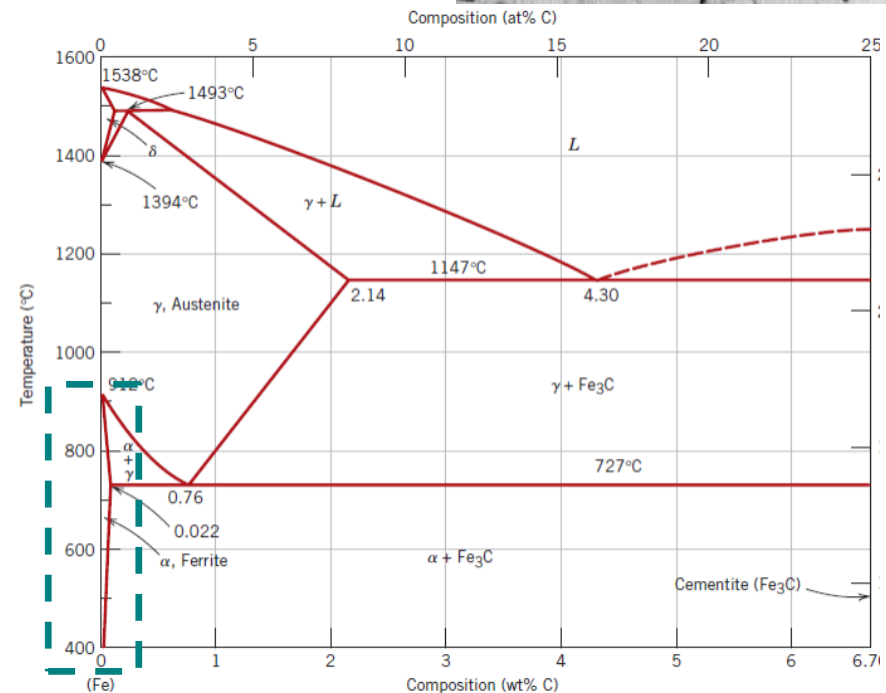
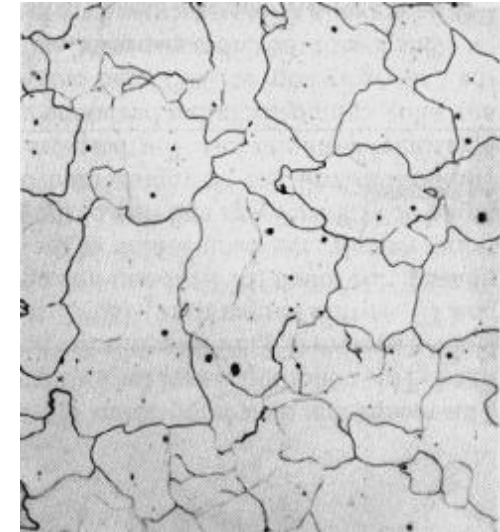
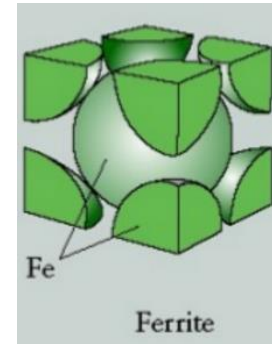
- 6.67 wt.% C
- Stoichiometry for Fe_3C
- Iron carbide or **cementite**
- This phase diagram is in fact Fe- Fe_3C

Fe-C SYSTEM

Phases in the Fe-C system

 α -Ferrite (BCC):

- Interstitial solid solution of C in **BCC iron**
- Low C concentration, maximum solubility **0.022 wt.% at 727°C**
- Soft phase
- Density 7.88 g/cm³

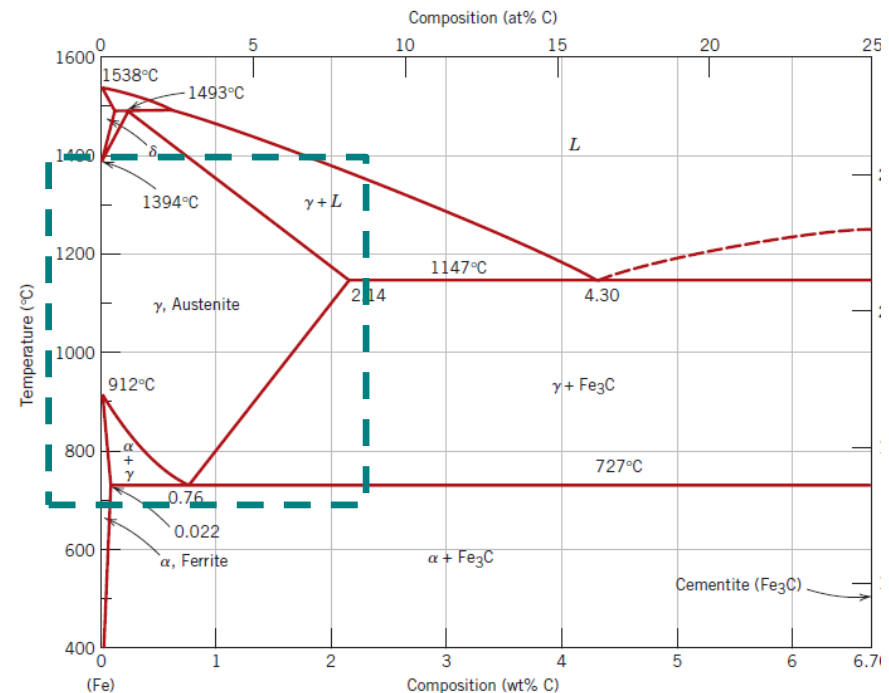
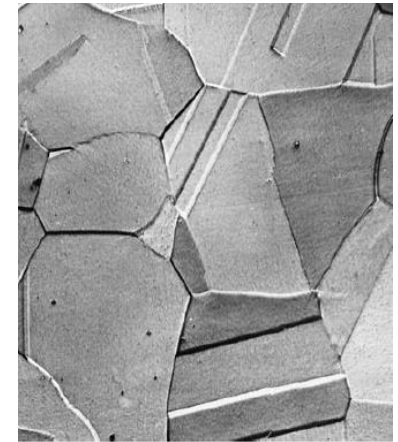
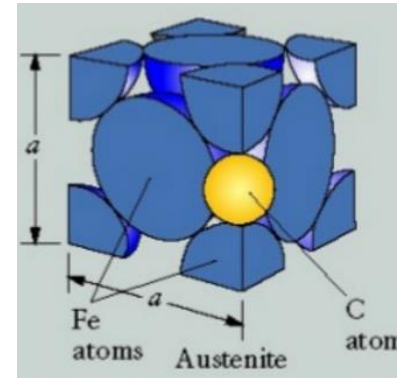


Fe-C SYSTEM

Phases in the Fe-C system

Austenite (FCC):

- interstitial solid solution of C in **FCC iron**
- Maximum solubility of C
2.14 wt.% at 1147 °C
- Important for phase transitions in heat treatment of steels
- Not magnetic

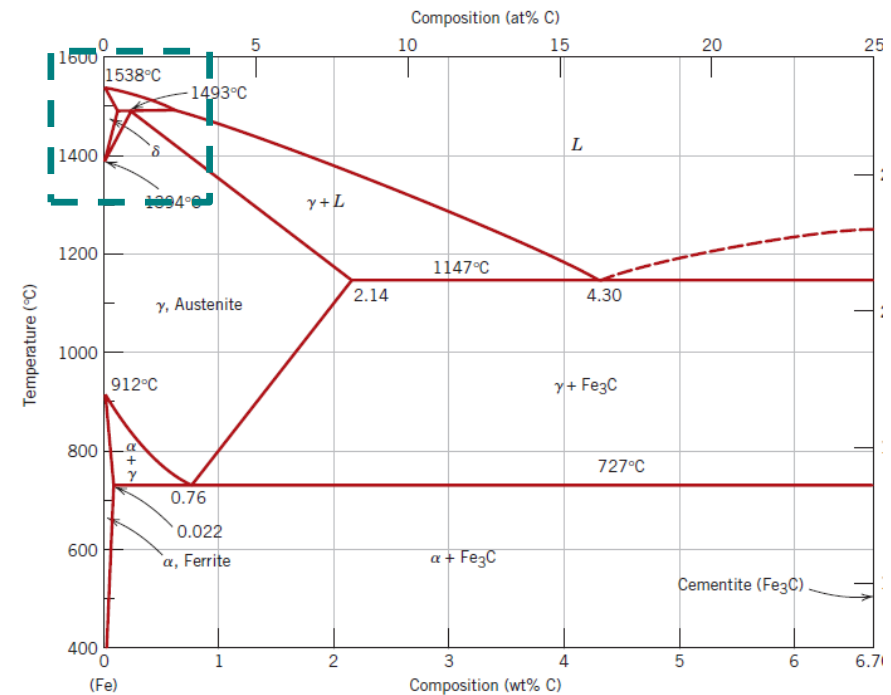


Fe-C SYSTEM

Phases in the Fe-C system

 δ -Ferrite (BCC):

- interstitial solid solution of C in **BCC iron**
- The same properties as α -Ferrite but at higher temperatures

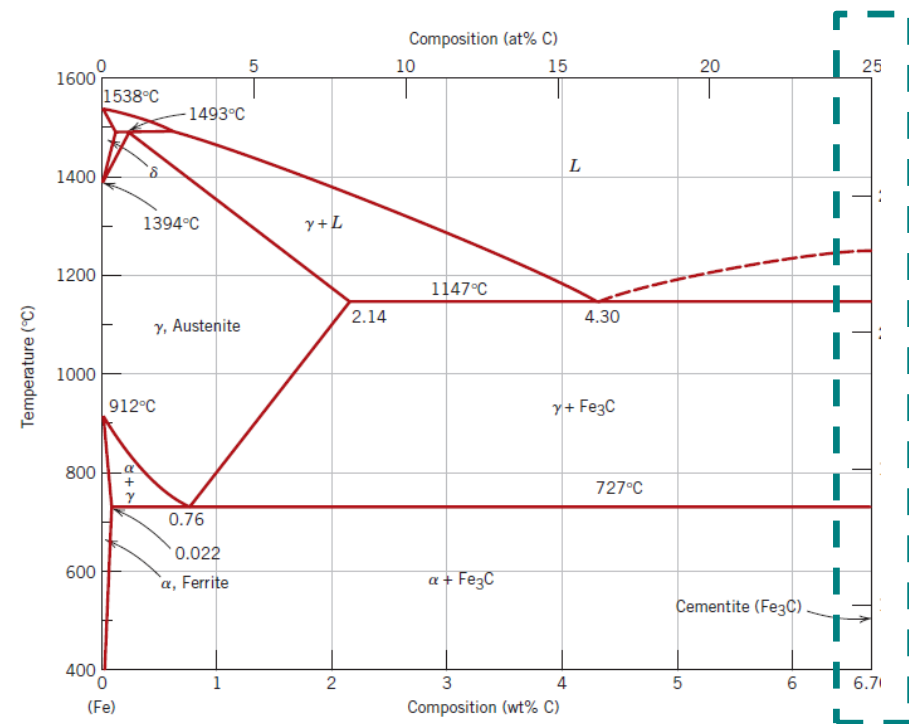
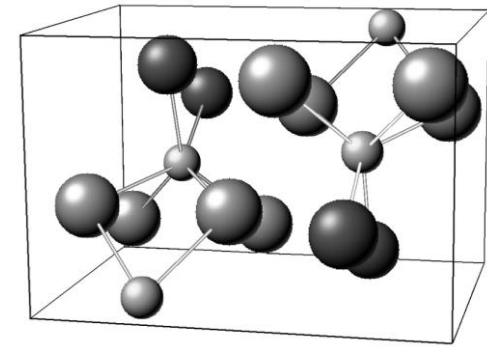


Fe-C SYSTEM

Phases in the Fe-C system

Cementite (Fe_3C):

- Intermetallic stoichiometric compound, 6.67 %C (Fe_3C)
- Forms when the solubility limit of carbon in α -Fe is exceeded below 727 °C
- Hard and brittle
- **Metastable compound**

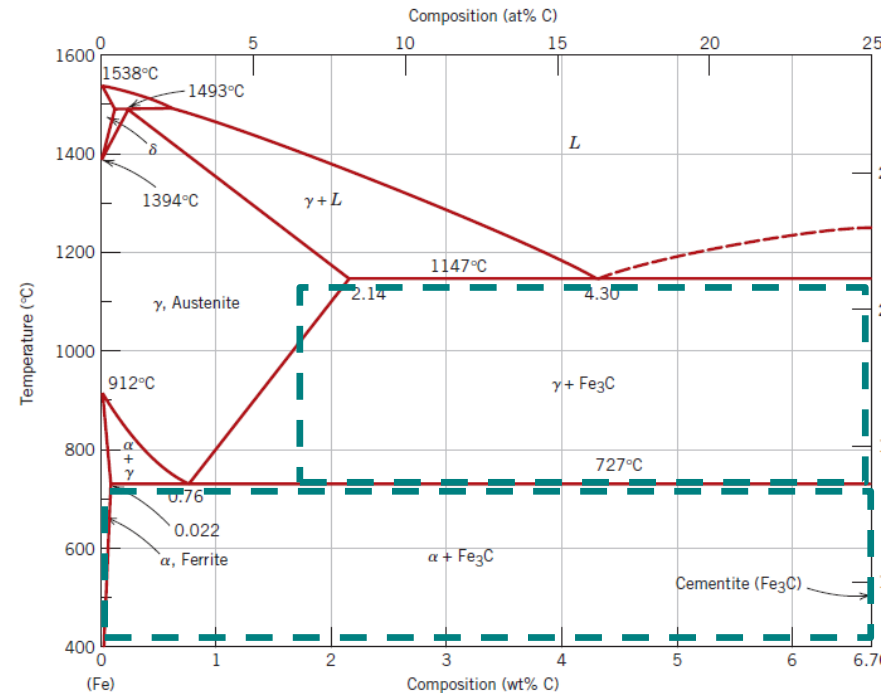


Fe-C SYSTEM

Phases in the Fe-C system

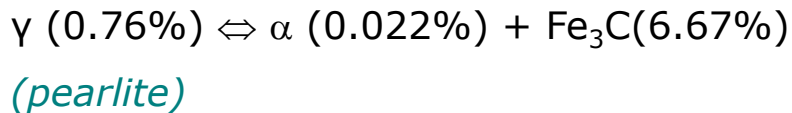
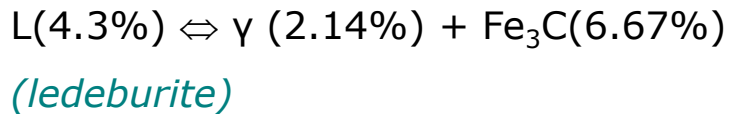
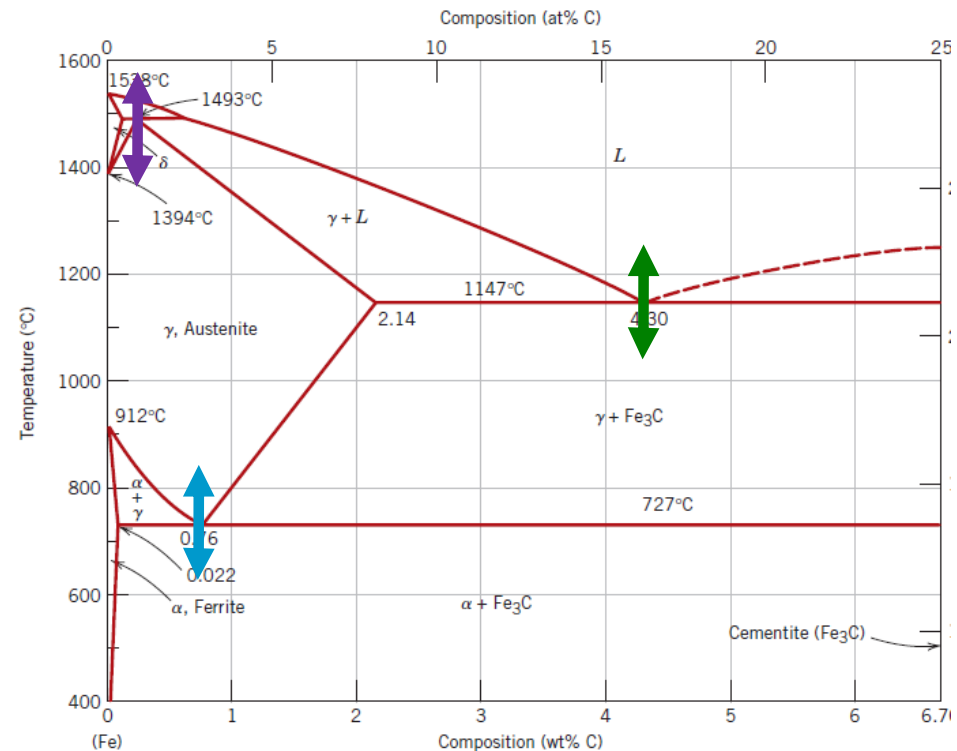
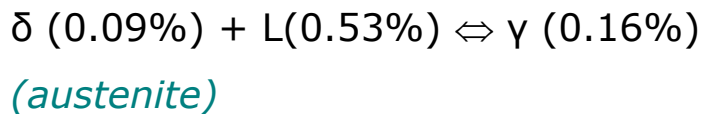
Two-phase regions:

- α -Ferrite + Fe_3C
Below 727 °C
- γ -Austenite + Fe_3C
Between 727°C and 1147°C



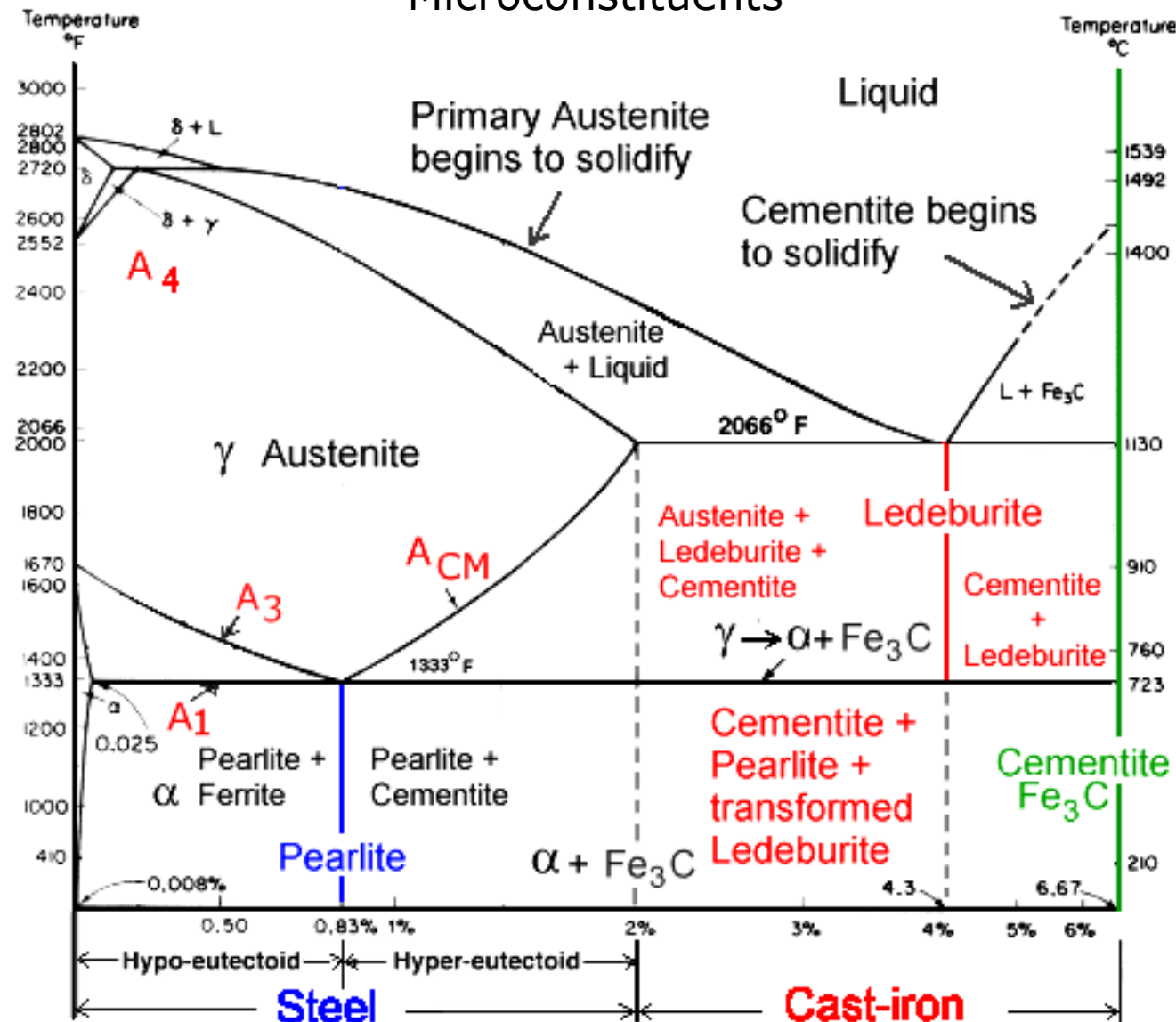
Fe-C SYSTEM

Invariant reactions in the Fe-C system

Eutectoid reaction (727°C):**Eutectic reaction (1147°C):****Peritectic reaction (1493°C):**

Fe-C SYSTEM

Microconstituents



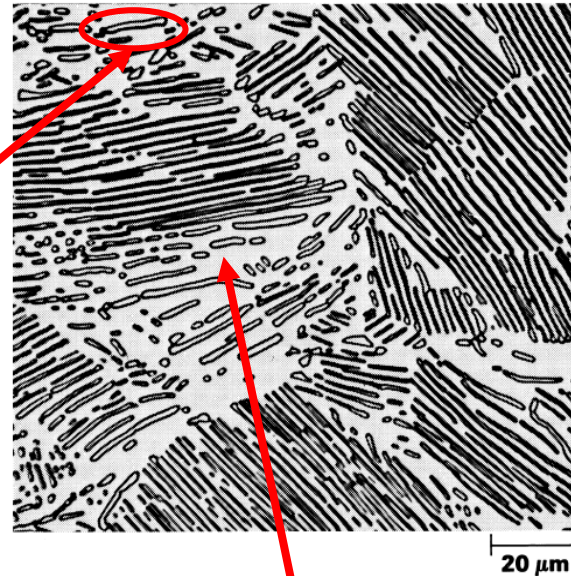
Fe-C: EQUILIBRIUM TRANSFORMATIONS

Eutectoid steels ($C_0 = 0.76\% \text{ C}$)

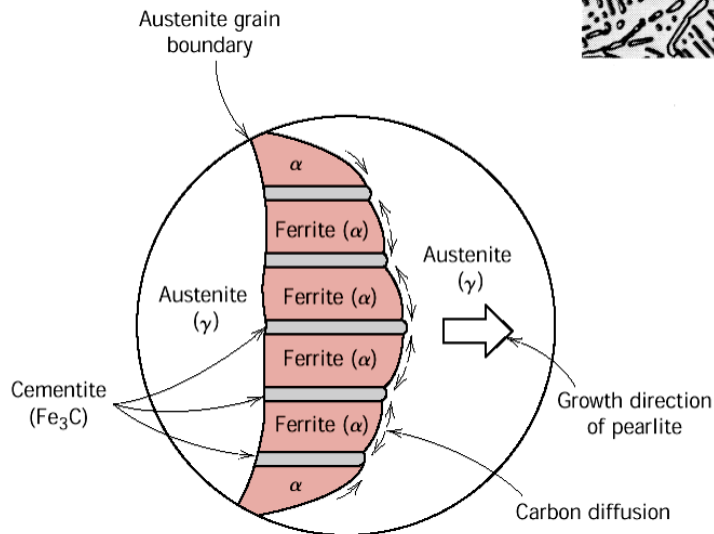
Pearlitic transformation

Only one microconstituent:
pearlite, formed by layers
(lamellas) of two phases: $\alpha + \text{Fe}_3\text{C}$

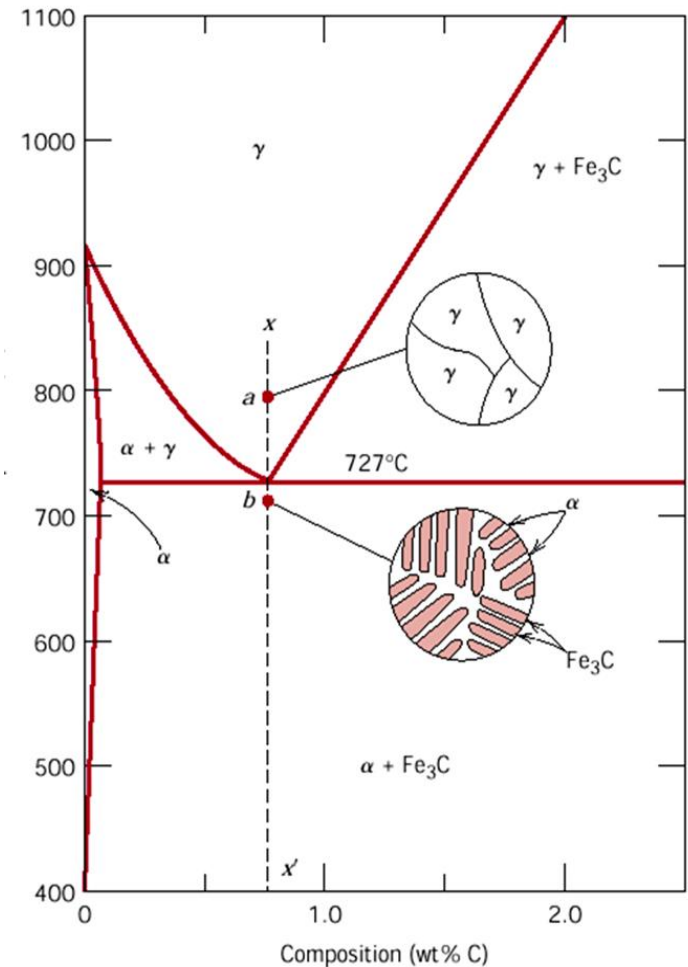
Cementite (Fe_3C)



Ferrite ($\alpha\text{-Fe}$)



Ferrite nucleates on the grain surface of austenite displacing C to adjacent zones through diffusion where cementite is formed.



Decreasing the pearlitic lamella spacing increases its hardness.

Fe-C: EQUILIBRIUM TRANSFORMATIONS

Eutectoid steels ($C_0 = 0.76\% \text{ C}$)

Information at point b

Microconstituents:

1 $\rightarrow$ Pearlite

Phases:

2 $\rightarrow$ Ferrite (α) y Cementite (Fe_3C)

Phase composition:

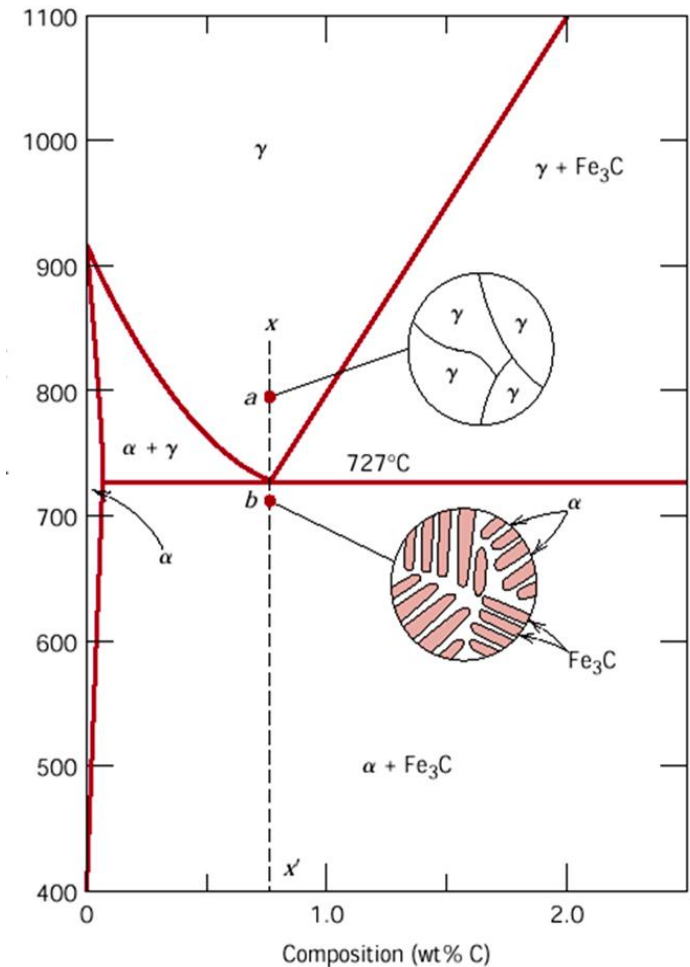
$$C_\alpha = 0.022\% \text{C}$$

$$C_{\text{Fe}_3\text{C}} = 6.67\% \text{C}$$

Phase amount:

$$W_\alpha = \frac{C_{\text{Fe}_3\text{C}} - C_0}{C_{\text{Fe}_3\text{C}} - C_\alpha} * 100 = \frac{6.67 - 0.76}{6.67 - 0.022} * 100 = 89\%$$

$$W_{\text{Fe}_3\text{C}} = 100 - 89 = 11\%$$



$\rightarrow$
6.67 %C

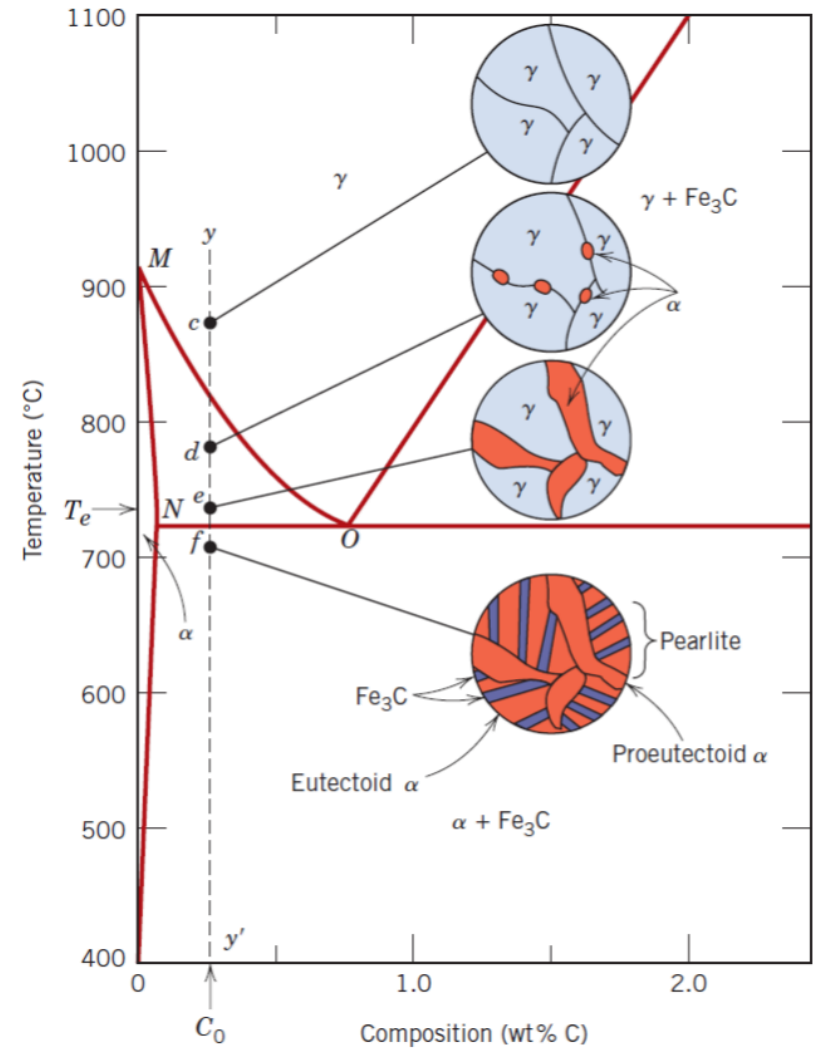
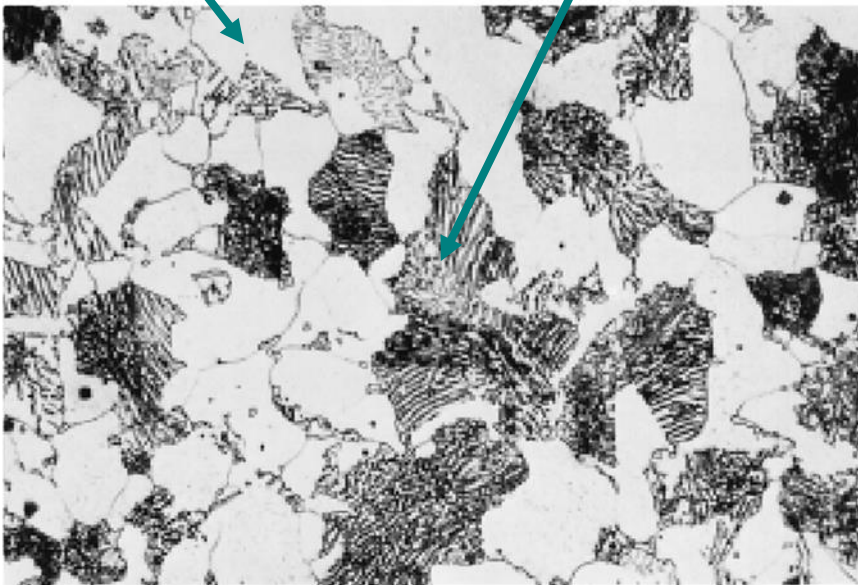
Fe-C: EQUILIBRIUM TRANSFORMATIONS

Hypoeutectoid steels ($C_0 < 0.76\% \text{ C}$)

Two microconstituents:
pearlite & proeutectoid ferrite

Proeutectoid ferrite or
primary (α_p)

Pearlite
($\alpha + \text{Fe}_3\text{C}$)



Fe-C: EQUILIBRIUM TRANSFORMATIONS

Hypoeutectoid steels ($C_0 < 0.76\% \text{ C}$)

$$C_0 = 0.3\% \text{ C}$$

Information at point c:

Phases:

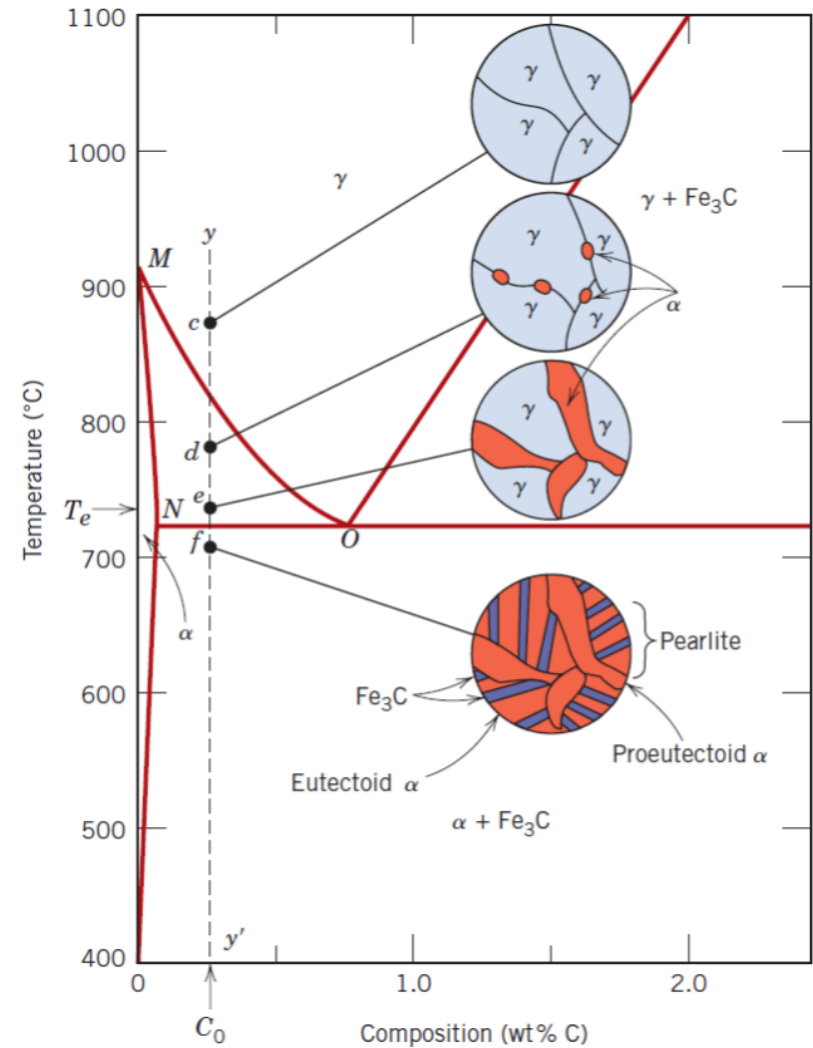
1 $\rightarrow$ Austenite (γ)

Phase composition:

$$C_\gamma = C_0 = 0.3\% \text{ C}$$

Phase amount:

$$W_\gamma = 100\%$$



Fe-C: EQUILIBRIUM TRANSFORMATIONS

Hypoeutectoid steels ($C_0 < 0.76\% \text{ C}$)

Information at point d:

Phases:

2 $\rightarrow$ Ferrite (α) y Austenite (γ)

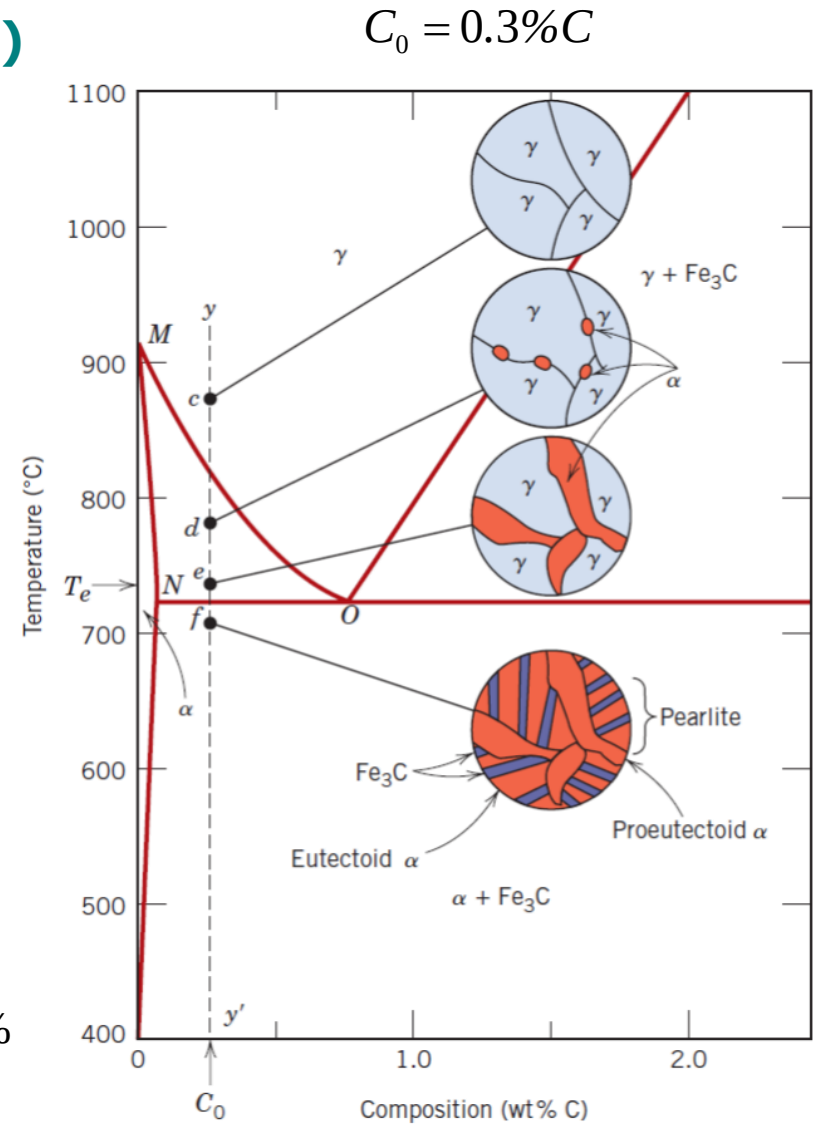
Phase composition:

$$C_\alpha = 0.02\% \text{C} \quad C_\gamma = 0.4\% \text{C}$$

Phase amount:

$$W_\alpha = \frac{S}{R+S} 100 = \frac{C_\gamma - C_0}{C_\gamma - C_\alpha} 100 = \frac{0.4 - 0.3}{0.4 - 0.02} 100 = 26\%$$

$$W_\gamma = \frac{R}{R+S} 100 = \frac{C_0 - C_\alpha}{C_\gamma - C_\alpha} 100 = \frac{0.3 - 0.02}{0.4 - 0.02} 100 = 74\%$$



Fe-C: EQUILIBRIUM TRANSFORMATIONS

Hypoeutectoid steels ($C_0 < 0.76\% \text{ C}$)

Information at point e:

Phases:

2 $\rightarrow$ Ferrite (α) y Austenite (γ)

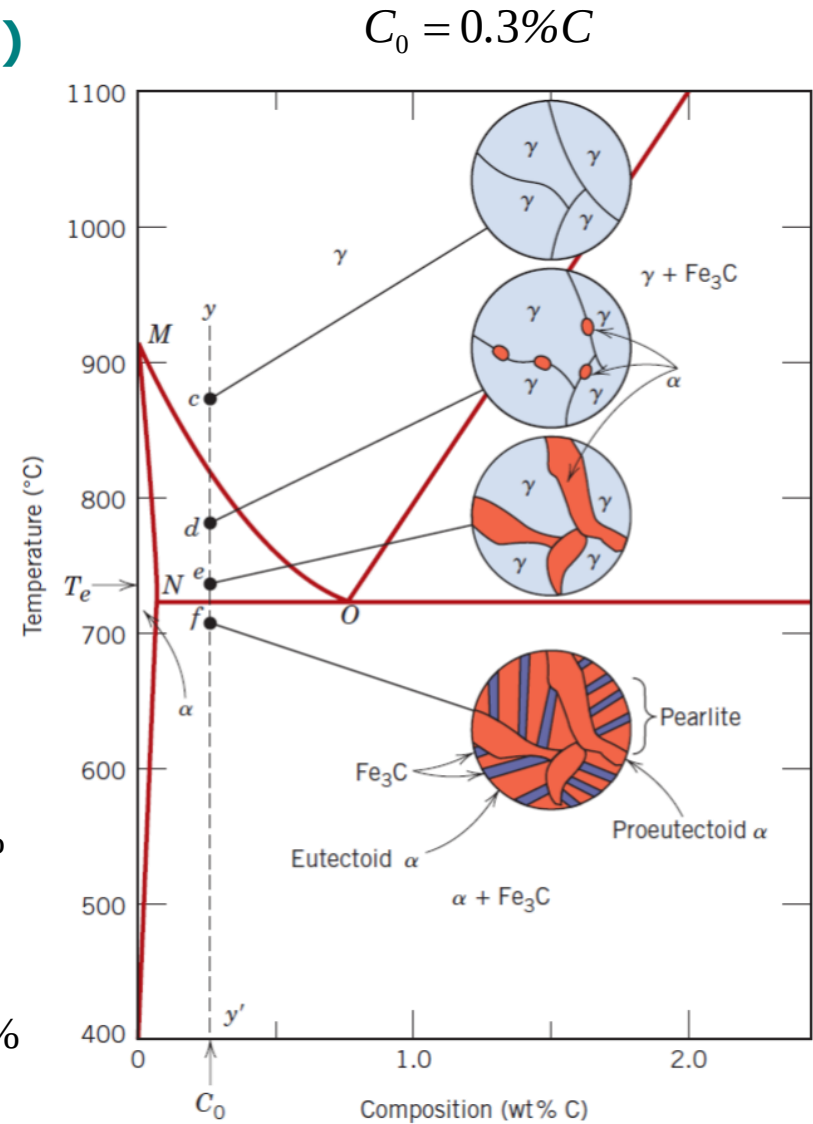
Phase composition:

$$C_\alpha = 0.02\% \text{ C} \quad C_\gamma = 0.76\% \text{ C}$$

Phase amount:

$$W_\alpha = \frac{S}{R+S} 100 = \frac{C_\gamma - C_0}{C_\gamma - C_\alpha} 100 = \frac{0.76 - 0.3}{0.76 - 0.02} 100 = 62\%$$

$$W_\gamma = \frac{R}{R+S} 100 = \frac{C_0 - C_\alpha}{C_\gamma - C_\alpha} 100 = \frac{0.3 - 0.02}{0.76 - 0.02} 100 = 38\%$$



Fe-C: EQUILIBRIUM TRANSFORMATIONS

Hypoeutectoid steels ($C_0 < 0.76\% \text{ C}$)

Information at point f:

Microconstituents: $2 \rightarrow \alpha_{\text{primary}}$ and Pearlite

Phases:

$2 \rightarrow \alpha$ ($\alpha_{\text{primary}} + \alpha_E$) and Cementite

Phase composition:

$$C_{\alpha} = 0.022\% \text{C} \quad C_{\text{Fe}_3\text{C}} = 6.67\% \text{C}$$

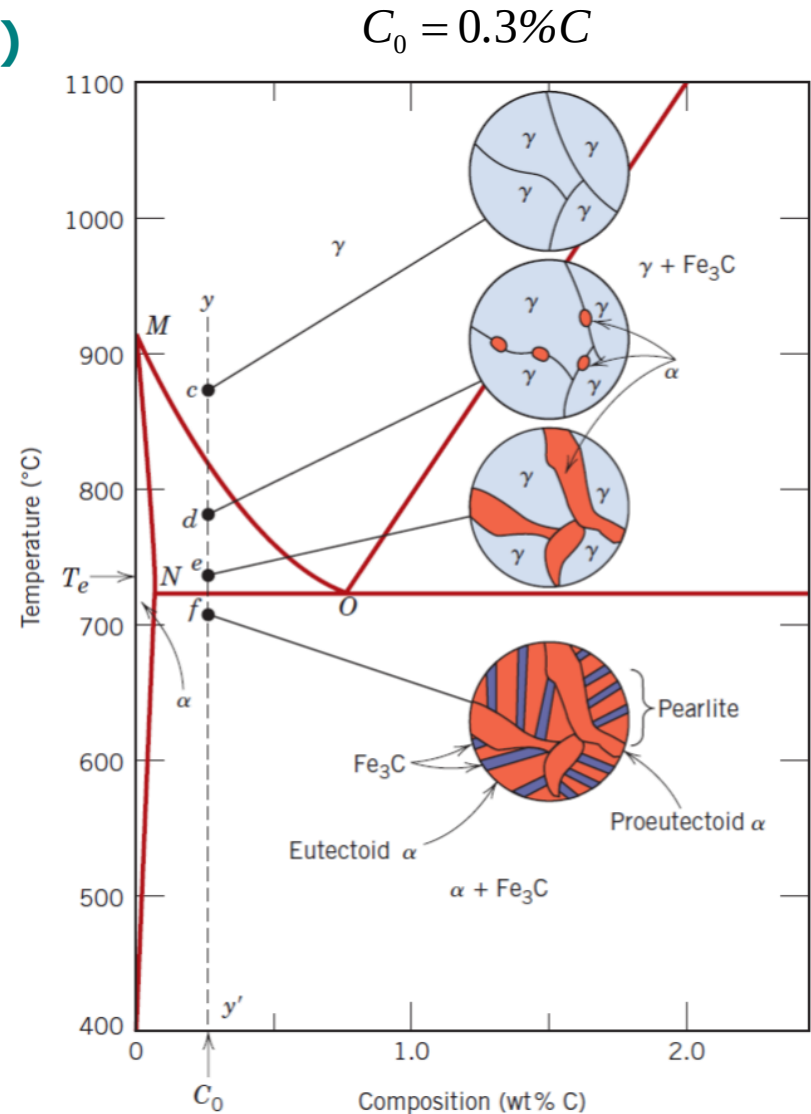
Phase amount:

$$W\alpha(\text{total}) = \frac{C_{\text{Fe}_3\text{C}} - C_0}{C_{\text{Fe}_3\text{C}} - C_{\alpha}} * 100 = \frac{6.67 - 0.3}{6.67 - 0.022} * 100 = 96\%$$

$$W\alpha_p = \frac{C_E - C_0}{C_E - C_{\alpha}} * 100 = \frac{0.76 - 0.3}{0.76 - 0.022} * 100 = 62\%$$

$$W\alpha_E = 96 - 62 = 34\%$$

$$W_{\text{Fe}_3\text{C}} = 100 - 96 = 4\%$$



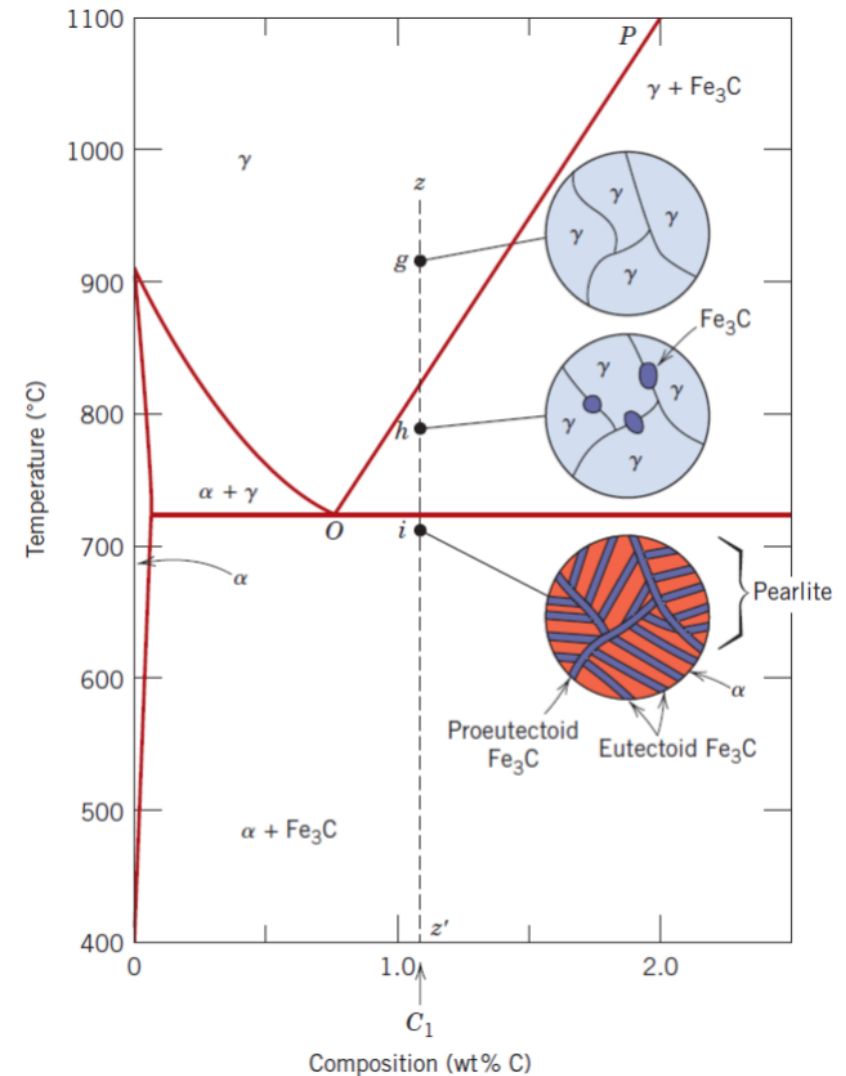
Fe-C: EQUILIBRIUM TRANSFORMATIONS

Hypereutectoid steels ($C_0 > 0.76\% \text{ C}$)

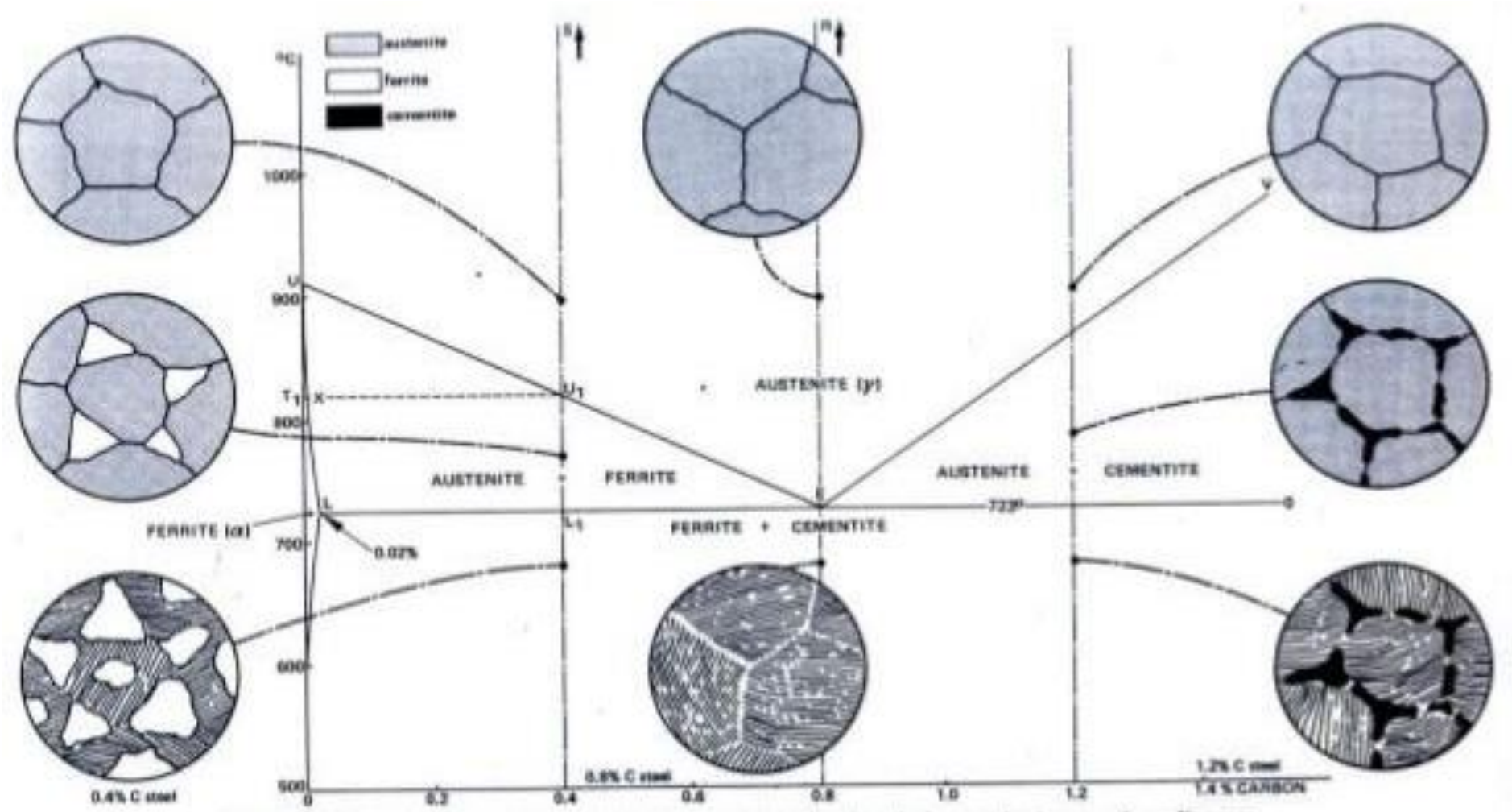
Two microconstituents:
pearlite & proeutectoid Fe_3C

Proeutectoid
cementite (Fe_3C)_P

Pearlite ($\alpha + \text{Fe}_3\text{C}$)



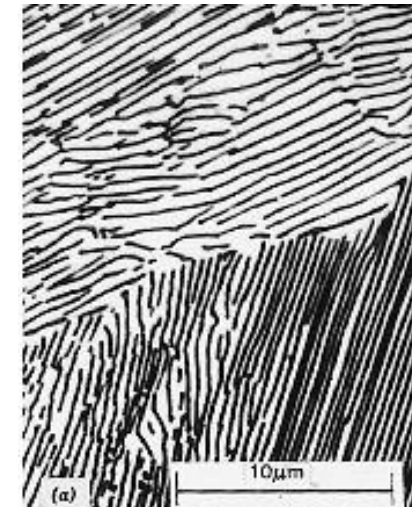
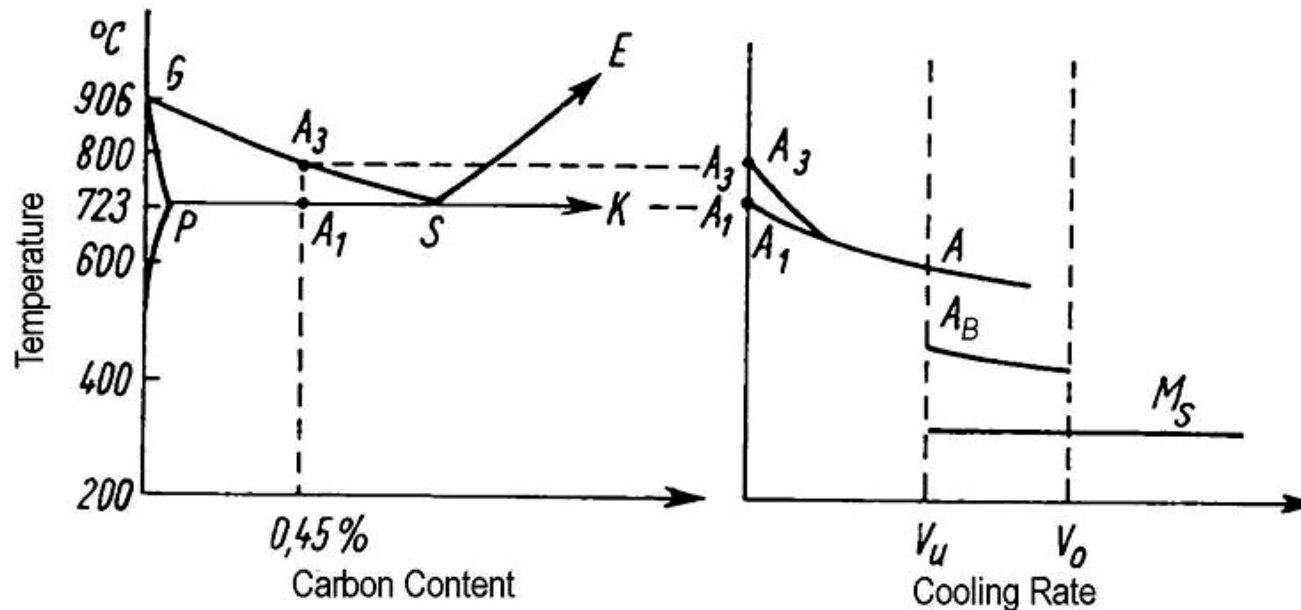
Fe-C: EQUILIBRIUM TRANSFORMATIONS



Fe-C: EQUILIBRIUM TRANSFORMATIONS

Pearlitic transformations

If the cooling rate increases a displacement of the eutectoid point is produced towards lower temperatures and lower C percentages



Coarse Pearlite



Fine Pearlite

Consequence:

Decreases the percentage of **proeutectoid ferrite (primary)** and the pearlitic lamella spacing increasing its hardness.

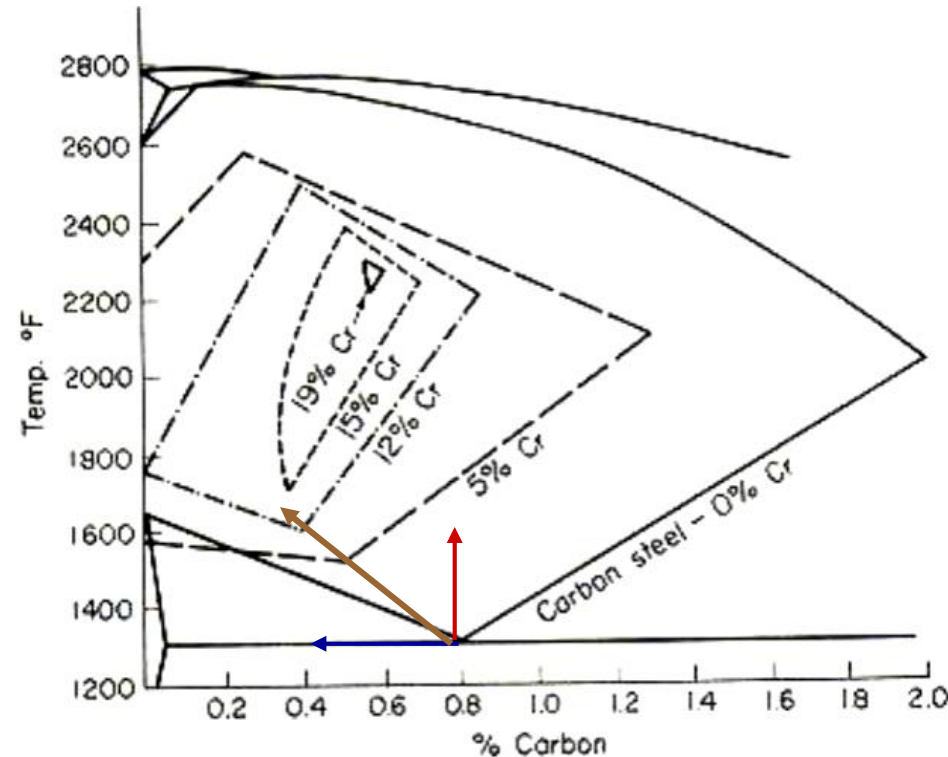
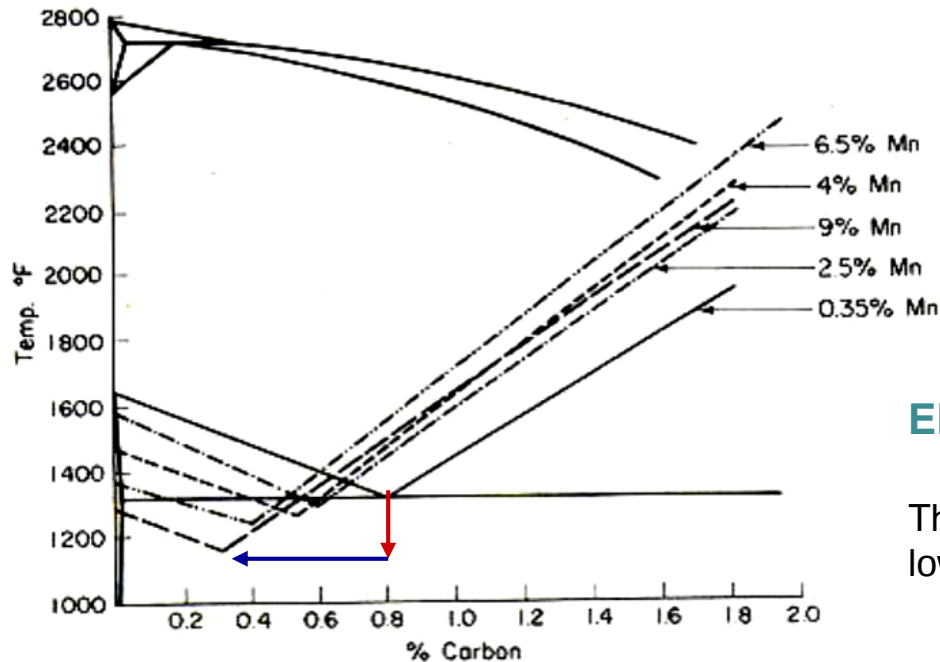
Decreasing the pearlitic lamella spacing increases its hardness.

INFLUENCE OF THE ALLOY ELEMENTS on Fe-C

Elements favouring α

They stabilize phase $\alpha \rightarrow$ displacing the eutectoid to higher T, at lower %C

Si, P, Al, Be, Sn, Sb, As,
Ti, Nb, V, Ta, Mo, W, Cr



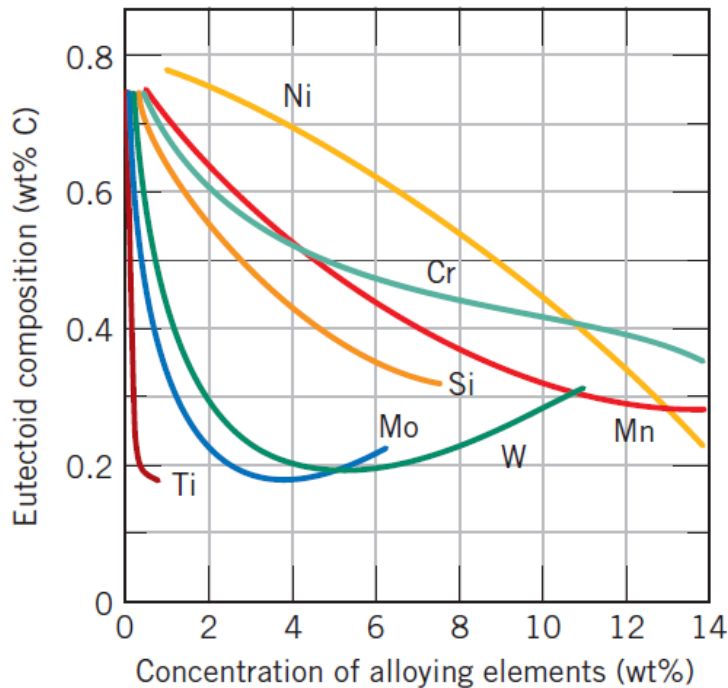
Elements favouring γ

They stabilize phase $\gamma \rightarrow$ displacing the eutectoid to lower T, at lower %C

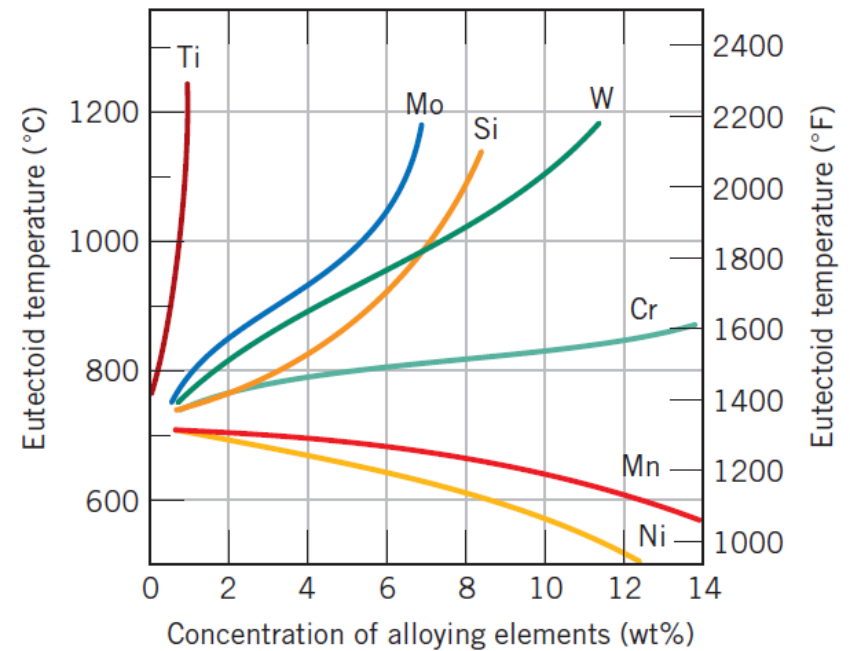
Mn, Co, Ni, Cu, Zn, Au

INFLUENCE OF THE ALLOY ELEMENTS on Fe-C METASTABLE

The presence of other elements in the Fe-C system can influence the temperature and composition at which the eutectoid reaction occurs.



Influence of alloying elements on the eutectoid composition (%C)



Influence of alloying elements on the eutectoid temperature